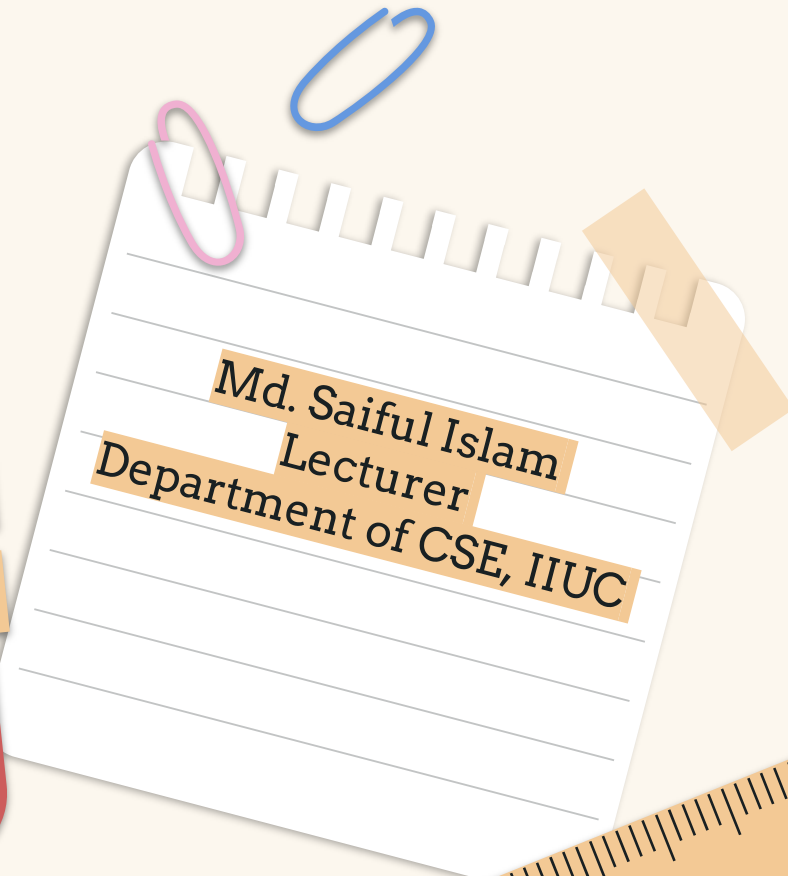


Discrete
Mathematics
Segment Two



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Introduction to Discrete Mathematics

Please see the video.

Logic

The rules of logic give precise meaning to mathematical statements. These rules are used to distinguish between valid and invalid mathematical arguments.

Our discussion begins with an introduction to the basic building blocks of logic—**propositions**.

Propositions

A **proposition** is a declarative sentence that is either true or false, but not both.

Example:

1. Washington, D.C., is the capital of the United States of America.
2. Toronto is the capital of Canada.
3. $1 + 1 = 2$.
4. $2 + 2 = 3$.

Propositions 1 and 3 are true, whereas 2 and 4 are false
All of the following declarative sentences are **propositions**.

Propositions

1. What time is it?
2. Read this carefully.
3. $x + 1 = 2$.
4. $x + y = z$.

Sentences 1 and 2 are not propositions because they are not declarative sentences. Sentences 3 and 4 are not propositions because they are neither true nor false.

Negation

Let p be a proposition. The *negation of p* , denoted by $\neg p$ (also denoted by $\overline{p}$), is the statement

“It is not the case that p .”

The proposition $\neg p$ is read “not p .” The truth value of the negation of p , $\neg p$, is the opposite of the truth value of p .

Example: Find the negation of the proposition
“Michael’s PC runs Linux”

Solution: The negation is
“Michael’s PC does not run Linux.”

TABLE 1 The Truth Table for the Negation of a Proposition.

p	$\neg p$
T	F
F	T

Conjunction

Let p and q be propositions. The *conjunction* of p and q , denoted by $p \wedge q$, is the proposition “ p and q .” The conjunction $p \wedge q$ is true when both p and q are true and is false otherwise.

Example: Find the conjunction of the propositions p and q where p is the proposition “Rebecca’s PC has more than 16 GB free hard disk space” and q is the proposition “The processor in Rebecca’s PC runs faster than 1 GHz.”

Solution: The conjunction of these propositions, $p \wedge q$, is the proposition “Rebecca’s PC has more than 16 GB free hard disk space, and the processor in Rebecca’s PC runs faster than 1 GHz.” This conjunction can be expressed more simply as “**Rebecca’s PC has more than 16 GB free hard disk space, and its processor runs faster than 1 GHz.**”

TABLE 2 The Truth Table for the Conjunction of Two Propositions.

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

Disjunction

Let p and q be propositions. The *disjunction* of p and q , denoted by $p \vee q$, is the proposition “ p or q .” The disjunction $p \vee q$ is false when both p and q are false and is true otherwise.

Example: Find the disjunction of the propositions p and q where p is the proposition “Rebecca’s PC has more than 16 GB free hard disk space” and q is the proposition “The processor in Rebecca’s PC runs faster than 1 GHz.”

Solution: The disjunction of p and q , $p \vee q$, is the proposition “*Rebecca’s PC has at least 16 GB free hard disk space, or the processor in Rebecca’s PC runs faster than 1 GHz*”.

TABLE 3 The Truth Table for the Disjunction of Two Propositions.

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

Implication

Let p and q be propositions. The *conditional statement* $p \rightarrow q$ is the proposition “if p , then q .” The conditional statement $p \rightarrow q$ is false when p is true and q is false, and true otherwise. In the conditional statement $p \rightarrow q$, p is called the *hypothesis* (or *antecedent* or *premise*) and q is called the *conclusion* (or *consequence*).

Example: Let p be the statement “Maria learns discrete mathematics” and q the statement “Maria will find a good job.” Express the statement $p \rightarrow q$ as a statement in English.

Solution: $p \rightarrow q$ represents the statement
“*If Maria learns discrete mathematics, then she will find a good job.*”

TABLE 5 The Truth Table for
the Conditional Statement
 $p \rightarrow q$.

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

Bi-implication

Let p and q be propositions. The *biconditional statement* $p \leftrightarrow q$ is the proposition “ p if and only if q .” The biconditional statement $p \leftrightarrow q$ is true when p and q have the same truth values, and is false otherwise. Biconditional statements are also called *bi-implications*.

Example: Let p be the statement “You can take the flight,” and let q be the statement “You buy a ticket.” Then $p \leftrightarrow q$ is the statement “***You can take the flight if and only if you buy a ticket.***”

TABLE 6 The Truth Table for the Biconditional $p \leftrightarrow q$.

p	q	$p \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

Tautology

A compound proposition that is always true, no matter what the truth values of the propositional variables that occur in it, is called a **tautology**.

Example: $p \vee \neg p$ is always true, it is a tautology.

p	$\neg p$	$p \vee \neg p$
T	F	T
F	T	T

Contradiction

A compound proposition that is always false is called a **contradiction**.

Example: $p \wedge \neg p$, is always false, it is a contradiction.

p	$\neg p$	$p \wedge \neg p$
T	F	F
F	T	F

Logically equivalent

The compound propositions p and q are called logically equivalent if $p \leftrightarrow q$ is a tautology.

The notation $p \equiv q$ denotes that p and q are logically equivalent.

Logically equivalent

Show that $p \rightarrow q$ and $\neg p \vee q$ are logically equivalent.

Solution: We construct the truth table for these compound propositions in Table 4. Because the truth values of $\neg p \vee q$ and $p \rightarrow q$ agree, they are logically equivalent. ◀

TABLE 4 Truth Tables for $\neg p \vee q$ and $p \rightarrow q$.

p	q	$\neg p$	$\neg p \vee q$	$p \rightarrow q$
T	T	F	T	T
T	F	F	F	F
F	T	T	T	T
F	F	T	T	T

Logically equivalent

TABLE 6 Logical Equivalences.

<i>Equivalence</i>	<i>Name</i>
$p \wedge \mathbf{T} \equiv p$ $p \vee \mathbf{F} \equiv p$	Identity laws
$p \vee \mathbf{T} \equiv \mathbf{T}$ $p \wedge \mathbf{F} \equiv \mathbf{F}$	Domination laws
$p \vee p \equiv p$ $p \wedge p \equiv p$	Idempotent laws
$\neg(\neg p) \equiv p$	Double negation law
$p \vee q \equiv q \vee p$ $p \wedge q \equiv q \wedge p$	Commutative laws

$(p \vee q) \vee r \equiv p \vee (q \vee r)$ $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$	Associative laws
$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$ $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$	Distributive laws
$\neg(p \wedge q) \equiv \neg p \vee \neg q$ $\neg(p \vee q) \equiv \neg p \wedge \neg q$	De Morgan's laws
$p \vee (p \wedge q) \equiv p$ $p \wedge (p \vee q) \equiv p$	Absorption laws
$p \vee \neg p \equiv \mathbf{T}$ $p \wedge \neg p \equiv \mathbf{F}$	Negation laws

Logically equivalent

Show that $\neg(p \vee (\neg p \wedge q))$ and $\neg p \wedge \neg q$ are logically equivalent by developing a series of logical equivalences.

Solution: We will use one of the equivalences in Table 6 at a time, starting with $\neg(p \vee (\neg p \wedge q))$ and ending with $\neg p \wedge \neg q$. (Note: we could also easily establish this equivalence using a truth table.) We have the following equivalences.

$\neg(p \vee (\neg p \wedge q))$	$\equiv \neg p \wedge \neg(\neg p \wedge q)$	by the second De Morgan law
	$\equiv \neg p \wedge [\neg(\neg p) \vee \neg q]$	by the first De Morgan law
	$\equiv \neg p \wedge (p \vee \neg q)$	by the double negation law
	$\equiv (\neg p \wedge p) \vee (\neg p \wedge \neg q)$	by the second distributive law
	$\equiv \mathbf{F} \vee (\neg p \wedge \neg q)$	because $\neg p \wedge p \equiv \mathbf{F}$
	$\equiv (\neg p \wedge \neg q) \vee \mathbf{F}$	by the commutative law for disjunction
	$\equiv \neg p \wedge \neg q$	by the identity law for $\mathbf{F}$

Consequently $\neg(p \vee (\neg p \wedge q))$ and $\neg p \wedge \neg q$ are logically equivalent.

Predicates

Predicates

Statements involving variables, such as

" $x > 3$,"

The statement " x is greater than 3" has two parts. The first part, the variable x , is the subject of the statement. The second part—the **predicate**, "is greater than 3"—refers to a property that the subject of the statement can have. We can denote the statement " x is greater than 3" by $P(x)$, where P denotes the predicate "is greater than 3" and x is the variable. The statement $P(x)$ is also said to be the value of the **propositional function** P at x . Once a value has been assigned to the variable x , the statement $P(x)$ becomes a proposition and has a truth value.

Predicates

EXAMPLE 1 Let $P(x)$ denote the statement “ $x > 3$.” What are the truth values of $P(4)$ and $P(2)$?

Solution: We obtain the statement $P(4)$ by setting $x = 4$ in the statement “ $x > 3$.” Hence, $P(4)$, which is the statement “ $4 > 3$,” is true. However, $P(2)$, which is the statement “ $2 > 3$,” is false. ◀

EXAMPLE 2 Let $A(x)$ denote the statement “Computer x is under attack by an intruder.” Suppose that of the computers on campus, only CS2 and MATH1 are currently under attack by intruders. What are truth values of $A(\text{CS1})$, $A(\text{CS2})$, and $A(\text{MATH1})$?

Solution: We obtain the statement $A(\text{CS1})$ by setting $x = \text{CS1}$ in the statement “Computer x is under attack by an intruder.” Because CS1 is not on the list of computers currently under attack, we conclude that $A(\text{CS1})$ is false. Similarly, because CS2 and MATH1 are on the list of computers under attack, we know that $A(\text{CS2})$ and $A(\text{MATH1})$ are true. ◀

Predicates

EXAMPLE 3 Let $Q(x, y)$ denote the statement " $x = y + 3$." What are the truth values of the propositions $Q(1, 2)$ and $Q(3, 0)$?

Predicates

EXAMPLE 3 Let $Q(x, y)$ denote the statement “ $x = y + 3$.” What are the truth values of the propositions $Q(1, 2)$ and $Q(3, 0)$?



Solution: To obtain $Q(1, 2)$, set $x = 1$ and $y = 2$ in the statement $Q(x, y)$. Hence, $Q(1, 2)$ is the statement “ $1 = 2 + 3$,” which is false. The statement $Q(3, 0)$ is the proposition “ $3 = 0 + 3$,” which is true. 

Quantifiers

Quantifiers

When the variables in a propositional function are assigned values, the resulting statement becomes a proposition with a certain truth value. However, there is another important way, called **quantification**, to create a proposition from a propositional function. Quantification expresses the extent to which a predicate is true over a range of elements. In English, the words *all*, *some*, *many*, *none*, and *few* are used in quantifications. We will focus on two types of quantification here: universal quantification, which tells us that a predicate is true for every element under consideration, and existential quantification, which tells us that there is one or more element under consideration for which the predicate is true. The area of logic that deals with predicates and quantifiers is called the **predicate calculus**.

Quantifiers

THE UNIVERSAL QUANTIFIER Many mathematical statements assert that a property is true for all values of a variable in a particular domain, called the **domain of discourse** (or the **universe of discourse**), often just referred to as the **domain**. Such a statement is expressed using universal quantification. The universal quantification of $P(x)$ for a particular domain is the proposition that asserts that $P(x)$ is true for all values of x in this domain. Note that the domain specifies the possible values of the variable x . The meaning of the universal quantification of $P(x)$ changes when we change the domain. The domain must always be specified when a universal quantifier is used; without it, the universal quantification of a statement is not defined.

Universal Quantifier

The *universal quantification* of $P(x)$ is the statement

“ $P(x)$ for all values of x in the domain.”

The notation $\forall x P(x)$ denotes the universal quantification of $P(x)$. Here $\forall$ is called the **universal quantifier**. We read $\forall x P(x)$ as “for all $x P(x)$ ” or “for every $x P(x)$.” An element for which $P(x)$ is false is called a **counterexample** of $\forall x P(x)$.

Universal Quantifier

EXAMPLE 8 Let $P(x)$ be the statement “ $x + 1 > x$.” What is the truth value of the quantification $\forall x P(x)$, where the domain consists of all real numbers?



Solution: Because $P(x)$ is true for all real numbers x , the quantification

$$\forall x P(x)$$

is true. 

EXAMPLE 9 Let $Q(x)$ be the statement “ $x < 2$.” What is the truth value of the quantification $\forall x Q(x)$, where the domain consists of all real numbers?

Solution: $Q(x)$ is not true for every real number x , because, for instance, $Q(3)$ is false. That is, $x = 3$ is a counterexample for the statement $\forall x Q(x)$. Thus

$$\forall x Q(x)$$

is false. 

Universal Quantifier

EXAMPLE 11

What is the truth value of $\forall x P(x)$, where $P(x)$ is the statement “ $x^2 < 10$ ” and the domain consists of the positive integers not exceeding 4?

Solution: The statement $\forall x P(x)$ is the same as the conjunction

$$P(1) \wedge P(2) \wedge P(3) \wedge P(4),$$

because the domain consists of the integers 1, 2, 3, and 4. Because $P(4)$, which is the statement “ $4^2 < 10$,” is false, it follows that $\forall x P(x)$ is false. 

EXAMPLE 12

What does the statement $\forall x N(x)$ mean if $N(x)$ is “Computer x is connected to the network” and the domain consists of all computers on campus?

Solution: The statement $\forall x N(x)$ means that for every computer x on campus, that computer x is connected to the network. This statement can be expressed in English as “Every computer on campus is connected to the network.” 

Existential Quantifier

THE EXISTENTIAL QUANTIFIER Many mathematical statements assert that there is an element with a certain property. Such statements are expressed using existential quantification. With existential quantification, we form a proposition that is true if and only if $P(x)$ is true for at least one value of x in the domain.

DEFINITION 2

The *existential quantification* of $P(x)$ is the proposition

“There exists an element x in the domain such that $P(x)$.”

We use the notation $\exists x P(x)$ for the existential quantification of $P(x)$. Here $\exists$ is called the *existential quantifier*.

Existential Quantifier

EXAMPLE 14 Let $P(x)$ denote the statement “ $x > 3$.” What is the truth value of the quantification $\exists x P(x)$, where the domain consists of all real numbers?



Solution: Because “ $x > 3$ ” is sometimes true—for instance, when $x = 4$ —the existential quantification of $P(x)$, which is $\exists x P(x)$, is true. 

EXAMPLE 15 Let $Q(x)$ denote the statement “ $x = x + 1$.” What is the truth value of the quantification $\exists x Q(x)$, where the domain consists of all real numbers?

Solution: Because $Q(x)$ is false for every real number x , the existential quantification of $Q(x)$, which is $\exists x Q(x)$, is false. 

Existential Quantifier

EXAMPLE 16 What is the truth value of $\exists x P(x)$, where $P(x)$ is the statement “ $x^2 > 10$ ” and the universe of discourse consists of the positive integers not exceeding 4?

Solution: Because the domain is $\{1, 2, 3, 4\}$, the proposition $\exists x P(x)$ is the same as the disjunction

$$P(1) \vee P(2) \vee P(3) \vee P(4).$$

Because $P(4)$, which is the statement “ $4^2 > 10$,” is true, it follows that $\exists x P(x)$ is true. 

Predicates and Quantifiers

EXAMPLE 23 Express the statement “Every student in this class has studied calculus” using predicates and quantifiers.

Solution: First, we rewrite the statement so that we can clearly identify the appropriate quantifiers to use. Doing so, we obtain:

“For every student in this class, that student has studied calculus.”



Next, we introduce a variable x so that our statement becomes

“For every student x in this class, x has studied calculus.”

Continuing, we introduce $C(x)$, which is the statement “ x has studied calculus.” Consequently, if the domain for x consists of the students in the class, we can translate our statement as $\forall x C(x)$.

Predicates and Quantifiers

EXAMPLE 24 Express the statements “Some student in this class has visited Mexico” and “Every student in this class has visited either Canada or Mexico” using predicates and quantifiers.

Solution: The statement “Some student in this class has visited Mexico” means that

“There is a student in this class with the property that the student has visited Mexico.”

We can introduce a variable x , so that our statement becomes

“There is a student x in this class having the property that x has visited Mexico.”

We introduce $M(x)$, which is the statement “ x has visited Mexico.” If the domain for x consists of the students in this class, we can translate this first statement as $\exists x M(x)$.

Predicates and Quantifiers

"Every student in this class has visited either Canada or Mexico"

"For every x in this class, x has the property that x has visited Mexico or x has visited Canada."

(Note that we are assuming the inclusive, rather than the exclusive, or here.) We let $C(x)$ be " x has visited Canada." Following our earlier reasoning, we see that if the domain for x consists of the students in this class, this second statement can be expressed as $\forall x(C(x) \vee M(x))$. However, if the domain for x consists of all people, our statement can be expressed as

"For every person x , if x is a student in this class, then x has visited Mexico or x has visited Canada."

In this case, the statement can be expressed as $\forall x(S(x) \rightarrow (C(x) \vee M(x)))$.

Predicates and Quantifiers

EXAMPLE 26 Consider these statements. The first two are called *premises* and the third is called the *conclusion*. The entire set is called an *argument*.

“All lions are fierce.”

“Some lions do not drink coffee.”

“Some fierce creatures do not drink coffee.”

(In Section 1.6 we will discuss the issue of determining whether the conclusion is a valid consequence of the premises. In this example, it is.) Let $P(x)$, $Q(x)$, and $R(x)$ be the statements “ x is a lion,” “ x is fierce,” and “ x drinks coffee,” respectively. Assuming that the domain consists of all creatures, express the statements in the argument using quantifiers and $P(x)$, $Q(x)$, and $R(x)$.

Solution: We can express these statements as:

$\forall x(P(x) \rightarrow Q(x)).$

if x is a lion then it is a fierce creature

$\exists x(P(x) \wedge \neg R(x)).$

there is at least one lion that do not drink coffee

$\exists x(Q(x) \wedge \neg R(x)).$

there is at least one fierce creature that do not drink coffee

Predicates and Quantifiers

EXAMPLE 27 Consider these statements, of which the first three are premises and the fourth is a valid conclusion.

“All hummingbirds are richly colored.”

“No large birds live on honey.”

“Birds that do not live on honey are dull in color.”

“Hummingbirds are small.”

Let $P(x)$, $Q(x)$, $R(x)$, and $S(x)$ be the statements “ x is a hummingbird,” “ x is large,” “ x lives on honey,” and “ x is richly colored,” respectively. Assuming that the domain consists of all birds, express the statements in the argument using quantifiers and $P(x)$, $Q(x)$, $R(x)$, and $S(x)$.

Predicates and Quantifiers

EXAMPLE 27 Consider these statements, of which the first three are premises and the fourth is a valid conclusion.

“All hummingbirds are richly colored.”

“No large birds live on honey.”

“Birds that do not live on honey are dull in color.”

“Hummingbirds are small.”

Let $P(x)$, $Q(x)$, $R(x)$, and $S(x)$ be the statements “ x is a hummingbird,” “ x is large,” “ x lives on honey,” and “ x is richly colored,” respectively. Assuming that the domain consists of all birds, express the statements in the argument using quantifiers and $P(x)$, $Q(x)$, $R(x)$, and $S(x)$.

Solution: We can express the statements in the argument as

$$\forall x(P(x) \rightarrow S(x)).$$

$$\neg \exists x(Q(x) \wedge R(x)). \quad \text{All large birds do not live on honey. } \forall x(Q(x) \rightarrow \neg R(x))$$

$$\forall x(\neg R(x) \rightarrow \neg S(x)).$$

$$\forall x(P(x) \rightarrow \neg Q(x)).$$

Exercise

Use quantifiers and predicates with more than one variable to express these statements.

- a) Every computer science student needs a course in discrete mathematics
- b) There is a student in this class who owns a personal computer.
- c) Every student in this class has taken at least one computer science course.
- d) There is a student in this class who has taken at least one course in computer science.
- e) Every student in this class has been in every building on campus.

Exercise

Solution:

- a) $\forall x N(x, \text{discrete mathematics})$, where $N(x, y)$ is “ x needs a course in y ” and the domain for x is computer science students and the domain for y is academic subjects
- b) $\exists x O(x, \text{personal computer})$, where $O(x, y)$ is “ x owns y ,” and the domain for x is students in this class
- c) $\forall x \exists y P(x, y)$, where $P(x, y)$ is “ x has taken y ”; x ranges over students in this class, and y ranges over computer science courses
- d) $\exists x \exists y P(x, y)$, with the environment of part (c) (i.e., the same definition of P and the same domain)
- e) $\forall x \forall y P(x, y)$, where $P(x, y)$ is “ x has been in y ”; x ranges over students in this class, and y ranges over buildings on campus

Exercise

Write each of these propositions in the form “p if and only if q” in English.

- a) If it is hot outside you buy an ice cream cone, and if you buy an ice cream cone it is hot outside.
- b) For you to win the contest it is necessary and sufficient that you have the only winning ticket.
- c) You get promoted only if you have connections, and you have connections only if you get promoted.
- d) If you watch television your mind will decay, and conversely.
- e) The trains run late on exactly those days when I take it

Answer:

- a) You buy an ice cream cone if and only if it is hot outside.
- b) You win the contest if and only if you hold the only winning ticket.
- c) You get promoted if and only if you have connections.
- d) Your mind will decay if and only if you watch television.
- e) The train runs late if and only if it is a day I take the train

Exercise

Let p and q be the propositions "The election is decided" and "The votes have been counted," respectively. Express each of these compound propositions as an English sentence.

- a) $\neg p$
- b) $p \vee q$
- c) $\neg p \wedge q$
- d) $q \rightarrow p$
- e) $\neg q \rightarrow \neg p$
- f) $\neg p \rightarrow \neg q$
- g) $p \leftrightarrow q$
- h) $\neg q \vee (\neg p \wedge q)$

Exercise

- a) $\neg p$: The election is not decided
- b) $p \vee q$: The election is decided or the votes have been counted
- c) $\neg p \wedge q$: The election is not decided and the votes have been counted
- d) $q \rightarrow p$: If the votes have been counted then the election is decided.
- e) $\neg q \rightarrow \neg p$: If the votes have not been counted then the election is not decided
- f) $\neg p \rightarrow \neg q$: If the election is not decided then the votes have not been counted
- g) $p \leftrightarrow q$: The election is decided if and only if the votes have been counted.
- h) $\neg q \vee (\neg p \wedge q)$: The votes have not been counted or, the election is not decided and the votes have been counted.

Exercise

Let p , q and r be the propositions:

p : Erik reads Newsweek

q : Erik reads The New Yorker

r : Erik reads Time

Express each of the following propositions as an English sentence:

- i. $(p \vee q) \wedge \neg r$
- ii. $(p \wedge q) \vee \neg(p \wedge r)$
- iii. $\neg(p \wedge \neg r)$
- iv. $\neg((r \vee q) \wedge \neg p)$
- v. $(p \rightarrow q) \vee (\neg q \wedge r)$
- vi. $(p \wedge q) \vee (\neg(q \wedge r))$
- vii. $\neg q \leftrightarrow r$

Exercise

Solution:

- i. $(p \vee q) \wedge \neg r$: Erik reads Newsweek or The New Yorker, but not Time
- ii. $(p \wedge q) \vee \neg(p \wedge r)$: Erik reads Newsweek and The New Yorker, or he does not read Newsweek and Time
- iii. $\neg(p \wedge \neg r)$: It is not true that Erik reads Newsweek but not Time
- iv. $\neg((r \vee q) \wedge \neg p)$: It is not true that Erik reads Time or the New Yorker but not Newsweek
- v. $(p \rightarrow q) \vee (\neg q \wedge r)$: If Erik reads Newsweek then he reads The New Yorker too or he does not read New Yorker but Time
- vi. $(p \wedge q) \vee (\neg(q \wedge r))$: Erik reads Newsweek and The New Yorker, or he does not read New Yorker and Time
- vii. $\neg q \leftrightarrow r$: Erik doesn't read the New Yorker if and only if he reads Time

Exercise

p: You have the flu.

q: You miss the final examination.

r: You pass the course.

i. $p \rightarrow q$

ii. $\neg q \leftrightarrow r$

iii. $q \rightarrow \neg r$

iv. $p \vee q \vee r$

v. $(p \rightarrow \neg r) \vee (q \rightarrow \neg r)$

vi. $(p \wedge q) \vee (\neg q \wedge r)$

Exercise

p: You have the flu.

q: You miss the final examination.

r: You pass the course.

i. $p \rightarrow q$: If you have flu then you will miss the final examination.

ii. $\neg q \leftrightarrow r$

iii. $q \rightarrow \neg r$

iv. $p \vee q \vee r$

v. $(p \rightarrow \neg r) \vee (q \rightarrow \neg r)$

vi. $(p \wedge q) \vee (\neg q \wedge r)$

Exercise

p: You have the flu.

q: You miss the final examination.

r: You pass the course.

- i. $p \rightarrow q$: If you have flu then you will miss the final examination.
- ii. $\neg q \leftrightarrow r$: You won't miss the final examination if and only if you pass the course.
- iii. $q \rightarrow \neg r$
- iv. $p \vee q \vee r$
- v. $(p \rightarrow \neg r) \vee (q \rightarrow \neg r)$
- vi. $(p \wedge q) \vee (\neg q \wedge r)$

Exercise

p: You have the flu.

q: You miss the final examination.

r: You pass the course.

- i. $p \rightarrow q$: If you have flu then you will miss the final examination.
- ii. $\neg q \leftrightarrow r$: You won't miss the final examination if and only if you pass the course.
- iii. $q \rightarrow \neg r$: If you miss the examination then you will be failing the course.
- iv. $p \vee q \vee r$
- v. $(p \rightarrow \neg r) \vee (q \rightarrow \neg r)$
- vi. $(p \wedge q) \vee (\neg q \wedge r)$

Exercise

p: You have the flu.

q: You miss the final examination.

r: You pass the course.

- i. $p \rightarrow q$: If you have flu then you will miss the final examination.
- ii. $\neg q \leftrightarrow r$: You won't miss the final examination if and only if you pass the course.
- iii. $q \rightarrow \neg r$: If you miss the examination then you will be failing the course.
- iv. $p \vee q \vee r$: You have the flu OR you miss the final examination OR you pass the course.
- v. $(p \rightarrow \neg r) \vee (q \rightarrow \neg r)$
- vi. $(p \wedge q) \vee (\neg q \wedge r)$

Exercise

p: You have the flu.

q: You miss the final examination.

r: You pass the course.

- i. $p \rightarrow q$: If you have flu then you will miss the final examination.
- ii. $\neg q \leftrightarrow r$: You won't miss the final examination if and only if you pass the course.
- iii. $q \rightarrow \neg r$: If you miss the examination then you will be failing the course.
- iv. $p \vee q \vee r$: You have the flu OR you miss the final examination OR you pass the course.
- v. $(p \rightarrow \neg r) \vee (q \rightarrow \neg r)$: If you have the flu then you'll not pass the course OR If you miss the final examination then you'll fail the course.
- vi. $(p \wedge q) \vee (\neg q \wedge r)$

Exercise

p: You have the flu.

q: You miss the final examination.

r: You pass the course.

- i. $p \rightarrow q$: If you have flu then you will miss the final examination.
- ii. $\neg q \leftrightarrow r$: You won't miss the final examination if and only if you pass the course.
- iii. $q \rightarrow \neg r$: If you miss the examination then you will be failing the course.
- iv. $p \vee q \vee r$: You have the flu OR you miss the final examination OR you pass the course.
- v. $(p \rightarrow \neg r) \vee (q \rightarrow \neg r)$: If you have the flu then you'll not pass the course OR If you miss the final examination then you'll fail the course.
- vi. $(p \wedge q) \vee (\neg q \wedge r)$: You have the flu and you miss the examination OR You will not miss the final examination and you pass the course.